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What is a picture worth? A history of remote sensing

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'Man must rise above the earth—to the top of the atmosphere and beyond—for only thus will he fully understand the world in which he lives.'

Socrates
500 BC

'If our ability to find and efficiently utilize resources does not accelerate, and accelerate rapidly, the industrial civilization we now enjoy will crumble within a few decades, for the economic status of any nation is almost always a direct function of the use it makes of available natural resources.

Two developments, aerial photography and airborne geophysical surveying techniques, have already increased the rate at which new knowledge of the world's resources can be acquired. But even with far wider use of the tools and techniques already available, the problems that face us are greater than our current ability to solve them. Fortunately, some of the further acceleration required can be obtained through use of remote sensing devices mounted in high flying aircraft and earth orbiting satellites.'

Dr William T. Pecora
Director, US Geological Survey, 1967

Abstract. Remote sensing is the use of electromagnetic energy to measure the physical properties of distant objects. It includes photography and geophysical surveying as well as newer techniques that use other parts of the electromagnetic spectrum. The history of remote sensing begins with photography. The origin of other types of remote sensing can be traced to World War II, with the development of radar, sonar, and thermal infrared detection systems. Since the 1960s, sensors have been designed to operate in virtually all of the electromagnetic spectrum. Today a wide variety of remote sensing instruments are available for use in hydrological studies; satellite data, such as Skylab photographs and Landsat images are particularly suitable for regional problems and studies. Planned future satellites will provide a ground resolution of 10–80 m.

Remote sensing is currently used for hydrological applications in most countries of the world. The range of applications includes groundwater exploration determination of physical water quality, snowfield mapping, flood-inundation delineation, and making inventories of irrigated land. The use of remote sensing commonly results in considerable hydrological information at minimal cost. This information can be used to speed-up the development of water resources, to improve management practices, and to monitor environmental problems.

Quelle est la valeur d'une image? Un tour d'horizon de télédétection

Résumé. La télédétection c'est l'emploi de l'énergie électromagnétique pour mesurer les propriétés

physiques des objets à distance. Elle comprend la photographie et la levée de plans géophysiques ainsi que des techniques plus récentes utilisant d'autres parties du spectre électromagnétique. L'histoire de la télédétection commence avec la photographie. L'origine d'autres types de télédétection remonte à l'époque de la Deuxième Guerre Mondiale, avec le développement des systèmes de détection radar, sonar et infrarouge thermique. Depuis les années de 1960 on a inventé des détecteurs pour commander virtuellement tout le spectre électromagnétique. Aujourd'hui une vaste diversité d'appareils de télédétection est disponible pour l'emploi dans les études hydrologiques; les données des satellites tels que les photographies de Skylab et les images de Landsat conviennent surtout aux problèmes et aux études régionaux. Pour les futurs satellites en cours, il a été prévu une résolution au sol de 10 à 80 m.

La télédétection actuelle a été utilisée pour les applications hydrologiques dans la plupart des pays du monde. L'étendue des applications comprend l'exploration des eaux souterraines, la détermination de la qualité physique de l'eau, mettre en carte des champs de neige, délimiter les inondations des crues, et l'inventaire des terrains irrigués. L'utilisation de la télédétection nous donne en général des informations hydrologiques considérables et à des frais minimaux. Cette information pourrait être utilisée afin d'accélérer le développement des ressources en eaux, améliorer la gestion, et mettre en surveillance les problèmes du milieu environnant.

INTRODUCTION

Remote sensing is the use of reflected and emitted energy to measure the physical properties of distant objects and their surroundings. Thus, remote sensing is a source of basic data, a science, and a tool. The measurements, whether recorded on a strip chart, magnetic tape, or film, are the data. Remote sensing is a science because the procedure consists of obtaining relative or absolute measurements, processing these data either manually or by machine, interpreting the results, and drawing meaningful conclusions. It is a tool because the conclusions can be used for making inventories of resources and for solving ecological problems.

Remote sensing includes the older sciences of photography, photogrammetry, and airborne geophysical surveying as well as newer techniques that use other parts of the electromagnetic spectrum.

Many sensors and techniques in remote sensing have application to hydrological problems and studies. This paper describes the development of remote sensing and current areas of hydrological research.

ORIGIN AND DEVELOPMENT

The history of remote sensing begins with photography, because this was the first sensor to be developed and perfected. Photography literally means to draw with light, and its history can be traced back about 2300 years to Aristotle's description of the camera obscura: light passing through a small aperture such as a pinhole will form an image (upside down) on the side of a darkened box or room. The camera obscura and later the camera lucida (still sold today as an aid to drawing) received some practical use in drawing landscapes, but mainly were items of curiosity.

Practical photography required a permanent method of fixing the image. Louis Daguerre of France accomplished this and reported his discovery in 1839 (Pollack, 1963, p. 35). The use of photography spread rapidly, and in 1847, 2000 cameras and half a million plates were sold in Paris alone (Pollack, 1963, p. 35). Improvements in

camera mechanisms, lenses, processing, and plates also were rapid. As a result, exposure time was shortened from the 15–30 min needed for Daguerre's initial method to 1/1000 s by 1875 (Pollack, 1963, p. 225). George Eastman introduced the first roll film and the first Kodak camera in 1888 (Pollack, 1963, p. 251).

The first plates and films were sensitive to only the blue and green parts of the visible spectrum (thus, a red safelight could be used in the darkroom with orthochromatic films). Colour sensitizers (called dyes) must be used to extend the range of silver halide films into the red and near infra-red. This principle was discovered by Vogel of Berlin in 1873 (Neblette, 1970, p. 14); in 1883, Abney detected a wavelength of 1.3 μm (in the near infra-red), using specially sensitized plates—a limit that still exists today (Hudson, 1969). Panchromatic film, however, which is sensitive to all of the visible spectrum, did not reach the commercial market until 1905, and black-and-white infra-red film was not used commercially until after World War II (Neblette, 1970, p. 18).

The theory of colour photography was worked out as early as 1868, but colour photography was impractical until 1930, when the Kodachrome process was introduced for movies (Pollack, 1963, p. 511). Film for still colour photos was introduced in 1938 but was not widely used; easy developing and processing were not possible until Ektachrome film became generally available about 1950 (Pollack, 1963, p. 511).

An early form of colour infra-red film was developed for camouflage detection late in World War II but was not used operationally (Fischer *et al.*, 1975). The three-layer colour infrared film in use today results mainly from developments during the Korean War in the early 1950s (Taylor & Mondey, 1972, p. 56).

The history of both aerial platforms for photography and photo interpretation are tied to intelligence requirements of the military in time of war. The earliest aerial photographs were taken from tethered balloons between 1850 and 1860; in 1862 the US Union Army used balloons to photograph or diagram (exact method uncertain) the defences around Richmond, Virginia (Avery, 1968). Similarly, Wilbur Wright made the first photographs from an airplane in 1909, only six years after Kitty Hawk (American Society of Photogrammetry, 1960, p. 6). Three years later, an Italian pilot obtained aerial photographs of military positions during a war with Turkey (Taylor & Mondey, 1972, p. 19).

Aerial photography and photo interpretation of a directed and practical nature, however, date from World War I. Before the end of the war, cameras had been designed specifically for aircraft use and had been mounted in the aircraft (for vertical photography and to free the hands of the observer). At times, French aerial units developed and printed more than 10 000 photographs per day; prints often were in the hands of army commanders within an hour of exposure (American Society of Photogrammetry, 1960, p. 5). In use by this time was the combination of panchromatic film and a yellow (minus blue) filter for haze penetration (Howard, 1970). In 1918, photo interpreters in the US First Army correctly detected and identified 90 per cent of the opposing German Army installations; their interpretation was verified after the armistice (American Society of Photogrammetry, 1960, p. 5). Scientific and commercial uses of aerial photography became important about 1930, and have increased steadily since that time. By 1940, hundreds of papers had been published on photo interpretation in the fields of archaeology, ecology, geology, pedology, forestry, engineering, and geography (American Society of

Photogrammetry, 1960, p. 7). Another large stimulus to photo interpretation was provided by military efforts in World War II and Korea. The US Army Air Force alone took 171 million negatives in World War II (Infield, 1970).

Photographic improvements in the 1950s consisted mainly of colour films and cameras designed for these films. During this same period, however, considerable progress was being made on instruments that operate outside the narrow visible part of the electromagnetic spectrum.

The origin of nonphotographic remote sensing can be traced to World War II, with the development and improvement of radar (radio detection and ranging), thermal infra-red detection systems, and sonar (sound navigation ranging). The airborne magnetometer and scintillometer were developed around the end of the 1940s. The need for military reconnaissance led to development and use of scanner systems including thermal infra-red and multispectral scanners, television, and SLAR (side-looking airborne radar) in the 1950s.

Many of the early military systems were being declassified in the 1960s. A few individuals at this time recognized the potential civilian application of these systems to resource problems. Thus, the first International Symposium on Remote Sensing of the Environment was held at the University of Michigan in the United States in 1962; it included papers on airborne geophysics, passive microwaves, radar imagery, and infra-red radiation. By 1964, the Michigan Symposium included papers on lasers, audio and radio frequency pulses, and a four-channel multispectral scanner.

Since 1964, sensors have been designed to operate in virtually all of the electromagnetic spectrum. Also, the resolution and capability of existing sensors have been improved greatly. A 24-channel multispectral scanner now exists that operates from the ultraviolet to the thermal infra-red, for example. Another development has been the tremendous expansion in earth-resource applications for remote sensing; most of this is due to the interest and sponsorship of the US National Aeronautics and Space Administration.

The development of platforms has progressed 'along with the development of instruments and applications. The advantages of high-altitude photographs for geological interpretations were recognized and described by W. R. Hemphill in 1958. The value of orbital photography was demonstrated in 1964 (Morrison & Chown) with geological sketch maps of the Sahara Desert, drawn from photographs obtained by the unmanned Mercury-4 spacecraft in 1961. The promising results obtained by multidisciplinary analysis and interpretation of the Mercury and Gemini photographs led in 1968 (Fischer & Robinove) to planning for the Earth Resources Technology Satellite (now called Landsat) series.

PRESENT AND FUTURE SYSTEMS

Today a wide variety of remote sensing instruments are available for use in hydrological studies. Gamma-ray spectrometers can be mounted in aircraft for soil moisture and snow-water-equivalent measurements. Similarly, aircraft-mounted television cameras and multispectral scanners can be used to measure reflected ultraviolet to near infra-red energy; thermal scanners detect emitted infra-red wavelengths; and SLAR and passive microwave radiometers probe the surface and near surface characteristics of materials at these wavelengths. For regional problems and

TABLE 1. Landsat images of the USA and tape-recorded images elsewhere in the world may be ordered from the EROS Data Center. Other images may be available for areas within 3000 km of the receiving stations listed below; such data must be ordered from the indicated agencies

Country	Receiving station	Agency address
Brazil	Cuiaba	Instituto De Pesquisas Espaciais (INPE) Attn: Divisao de Banco de Dados, Av. Dos Astronautas 1758, Caixa Postal 515, 12.200-Sao Jose dos Campos, SP Brasil
Canada	Prince Albert	Integrated Satellite Information, Box 1630, Prince Albert, Saskatchewan, Canada S6U 5T2
Canada	St John's	Department of Forestry and Agriculture, Building 810, Pleasantville, St John's, Newfoundland, Canada A1A 1P9
Iran	Shahdasht	Iranian Remote Sensing Center, National Iranian Radio and Television, 80 West Sepand Avenue, Tehran, Iran
Italy	Fucino	Telespazio—SES, Corso d'Italia, L-3 Roma, Italy
World	—	EROS Data Center, Sioux Falls, South Dakota 57198, USA

studies, satellite data such as Skylab photographs and Landsat images are available for analysis and interpretation.

Landsat images have been acquired every 9–18 days on a repetitive basis since mid 1972. At least several images are available nearly everywhere in the world.* These data can be obtained by the public from the sources listed in Table 1. The MSS (multispectral scanner) on Landsats 1, 2, and 3 obtains images at green, red and near infra-red wavelengths and has an instantaneous field-of-view (nominal resolution) of 79 m. The RBV (return beam vidicon) television cameras on Landsats 1 and 2 were little used but produced a few images at similar wavelengths and resolution.

Landsat 3, launched in early 1978, contains the MSS system plus a thermal infra-red channel (band 8) and a new RBV system. Part of the thermal channel failed after producing only a few images, but the other sensors are operating as planned. The new RBV produces black-and-white panchromatic images with a nominal resolution of 30 m; these images are degraded by atmospheric haze but have excellent quality when the atmosphere is clear.

Seasat and HCMM (Heat Capacity Mapping Mission) satellites also were launched in early 1978. The imaging SAR (synthetic-aperture radar) on Seasat and the thermal infra-red scanner on HCMM are of interest for both terrestrial and oceanic hydrology. The SAR† is L-band (25 cm wavelength) and has a slight penetration capability for thin vegetation, dry soil, dry snow, and ice; resolution is about 25–40 m, depending on the method of data processing; repeat cycle for exact coverage is 152 days. The HCMM thermal scanner operates at 10.5–12.5 μm wavelength, has a ground resolution of 500 m, and repeats coverage every 1.5–3.5 days. Unfortunately, these satellites do not have an image tape recording capability,

* Landsat data generally are acquired only over a large land masses. Also, the acquisition of data in many areas has been limited by cloud cover and capacity of the tape recording system.

† Seasat had an electrical failure in October 1978, and only a few strips of imagery were obtained over land areas by the SAR.

TABLE 2. Seasat and HCMM satellite images are available only for areas within about 1500 km radius of ground receiving stations

<i>HCMM (Heat Capacity Mapping Mission)</i>	
Country	Receiving station
Australia	Orroral
Canada	Shoe Cove, Newfoundland
Spain	Madrid
USA	Fairbanks, Alaska
	Goldstone, California
	Rosman, North Carolina

For additional information on data availability, contact: National Space Science Data Center, Code 601, Goddard Space Flight Center, Greenbelt, Maryland 20771, USA

<i>Seasat</i>	
Country	Receiving station
Britain	Oakhanger, England*
Canada	Shoe Cove, Newfoundland
Spain	Canary Islands*
USA	Fairbanks, Alaska
	Goldstone, California
	Merrit Island, Florida

For further information on data availability, contact: National Environmental Satellite Service, NOAA, Iverson Mall, 3737 Branch Avenue, Marlow Heights, Maryland 20031, USA

* Under negotiation with the European Space Agency.

and data are available only within about 1500 km radius from the ground receiving stations listed in Table 2.

In the future, Space Shuttle Mission OFT-2 (late 1979) may carry a modified Seasat SAR Shuttle Mission STS-9 (about August 1980) will carry a 30.5 cm focal length, metric camera of French design; this camera will provide 10–30 m resolution (depending on film type); orbital constraints, however, will restrict coverage to latitudes 30°N–30°S. Spacelab, under development by the European Space Agency, will be a multipurpose experiment that will begin with a Shuttle launch in 1981; A Zeiss 30/23 metric camera will produce photographs with a resolution of about 20 m. In about 1983, the French plan the launch of SPOT satellite, which will use a MLA (multiple linear array) of solid state detectors to achieve a ground resolution of 10–20 m. There also are tentative proposals for a German ARGUS spacecraft and five JEOS (Japanese Earth Observation Satellites) with visible and near infrared sensors capable of 15–50 m resolution.

Landsat D, scheduled for launch in early 1981, will include the same 4-band MSS

TABLE 3. There are many operational and experimental uses of remote sensing for studies and problems in terrestrial hydrology

Application	Operational (O) or experimental (E)
<i>Surface water</i>	
1. Inventory of lakes and ponds	O
2. Flood wave delineation	O
3. Inventory of floating vegetation	O
4. Delineation of drainage patterns	O
5. Detection of oil slicks	O
6. Dye time-of-travel measurements	O
7. Monitor glacier advance or recession	O
8. Wetlands delineation	O
9. Snowfield mapping	O/E
10. Inventory of irrigated land	O/E
11. Shoreline and river morphology studies	O/E
12. Estimates of transpiration, based on vegetation area and volume	O/E
13. Inventory of springs and seeps, including underwater and shoreline springs	O/E
14. Relationship of drainage patterns to lithology, structure, and geomorphological history	O/E
15. Estimates of land-surface permeability and percentage of impervious area	O/E
16. Inventory of underwater vegetation	O/E
17. Calculation of water depth	O/E
18. Delineation and monitoring current patterns in rivers, lakes, and estuaries	O/E
19. Relationship of changing land use to drainage characteristics	E
20. Estimates of water volume in lake and reservoir storage	E
21. Relationship of land cover to streamflow statistics	E
22. Delineation of flood limits after flood recession	E
23. Relationship of saturated-soil area in drainage basin to amount and source of streamflow	E
24. Estimates of snow thickness, water equivalent, and free-water content	E
<i>Groundwater: surface and near surface aquifers</i>	
1. Delineation of aquifer boundaries	O
2. Inference of primary and secondary porosity and permeability	O
3. Estimates of water table depth and saturated thickness	O
4. Delineation of probable recharge and discharge areas; relationship between topography and hydrology	O
5. Inference of relationship between groundwater and surface water	O
6. Estimates of well production	O
7. Interpretation of groundwater quality; fresh or saline	O
<i>Quality of water</i>	
1. Measurement of water temperature	O
2. Measurement of water colour, turbidity and productivity	O/E
3. Monitor lake and estuary eutrophication	O/E
4. Determine source, extent, and dispersion of pollutants	O/E
5. Measurement of fluorescing pollutants	O/E
6. Estimation of dissolved chemical concentrations	E

Operational (O): Remote sensing is presently used to obtain this type of hydrological information.

Partly operational (O/E): Some information is presently obtained; other aspects are in a research or testing phase of development.

Experimental (E): Techniques are in a research phase of development.

on Landsats 1–3 plus a 7-band multispectral scanner called the Thematic Mapper (TM). Resolution on the TM will be about 30 m; the repeat cycle is still under discussion.

Two other proposed future satellites are Stereosat and SEOS (Synchronous Earth Observation Satellite). Both proposals call for use of an MLA to achieve a resolution of 20–60 m. Stereosat would use two telescopes at a 60° angle to obtain stereo coverage.

APPLICATIONS

Remote sensing is currently used on an operational or experimental basis for hydrological applications in most countries of the world. The range of applications includes surface water, groundwater, and water quality studies (Table 3). Remote sensing should be considered as a tool to be used for these studies, however, and not as a solution to the problems.

The advantages of remote sensing may best be shown by an application that presently is widely used. For groundwater exploration, remote sensing is a fast, inexpensive method of obtaining some information on surface and near surface aquifers (Table 3). Specifically, by including the interpretation of aerial photographs and satellite images in an exploration programme, there is less need for field work and less need for other, more expensive exploration methods. New or better geological and hydrological information also might be obtained from remote sensing data; this information may aid or confirm the interpretation of data obtained by other methods. Finally, remote sensing is the only practical method of recording transitory soil moisture and vegetation patterns (which may be indicators of groundwater occurrence or quality) over large areas.

The accuracy of results obtained by interpretation of remote sensing data depends on the knowledge, experience, and skill of the scientist. So do results of all other groundwater exploration methods. A consideration of available tools shows that all groundwater exploration methods depend on interpretation. A geologist infers, for example, the presence, depth, and attitude of lithological units between outcrops and test wells. Also, a geologist interprets geophysical logs and surveys. At best, a geological or hydrological map shows the most logical and likely interpretation of all available facts and evidence.

Remote sensing also has similar advantages and promise in surface water and water quality applications (Table 3). In most cases, remote sensing should be used for hydrological studies and research only as a supplement to conventional procedures. Nevertheless, the use of remote sensing may add considerable hydrological information at minimal cost. This information can be used for water resource development, environmental monitoring, and improving management practices.

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